

Mahdi Farrokhmaleki

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SUMMARY

Machine Learning and Software Engineer with an M.Sc. in Computer Science, focused on data-driven systems and applied AI. Experienced in building end-to-end machine learning pipelines and user-facing applications across web and cloud platforms, with strong emphasis on scalability, performance, and maintainability.

TECHNICAL SKILLS

Programming Languages: Python, C++, C#, Java, JavaScript, SQL

ML and Applied AI: Computer Vision, Reinforcement Learning, NLP, LLMs, LangChain

Web and Services: React, FastAPI, REST APIs, Gradio

Data and Libraries: NumPy, Pandas, OpenCV, Matplotlib, PyTorch, TensorFlow, Scikit-learn, Selenium

Databases: MySQL, PostgreSQL, PostGIS, MongoDB

Cloud and DevOps: Docker, CI/CD, GitHub Actions, AWS (S3, EC2, Textract, Bedrock), MLflow, Kubernetes

Tools and Platforms: Linux, Git, Unity, Unreal Engine, Tableau, Unity ML-Agents, Blender, ArcGIS

EXPERIENCE

Research Assistant

Multisensory Studio (MSS), University of Calgary

Oct 2025 to Present

Calgary, Canada

- Developed and deployed full-stack web applications.
- Designed and implemented backend APIs to support application functionality and data flow.
- Built and managed user databases, enabling secure storage and efficient data retrieval.
- Created interactive web interfaces to support user interaction.
- Leveraged AWS services and large language models (LLMs) to automate document processing workflows, increasing the speed of legal case handling.

Research Assistant

Serious Games Research Group

Sep 2023 to Apr 2026

Calgary, Canada

- Designed and implemented an automated PCG repair pipeline using Python, C#, and machine learning models.
- Developed ML-based techniques to detect and correct structural instabilities in generated game content, improving generation efficiency by 45%.
- Publications: [AIIDE 2025](#), [AIIDE 2024](#), [AVI 2024 Workshop](#).

GIS R&D Intern

City of Calgary

May 2024 to Sep 2025

Calgary, Canada

- Worked as part of an R&D and technical team on Digital Twin systems and their real-world applications.
- Developed simulation and analysis tools using ArcGIS to support urban data insights.
- Generated 3D models from aerial imagery and integrated them into ArcGIS and Unity for visualization.
- Worked with city-scale databases to process and manage geospatial data.
- Built full-stack web applications to visualize city data for cross-departmental use.

Research Assistant

APA AI Lab, University of Tabriz

Feb 2022 to Aug 2023

Tabriz, Iran

- Conducted research on policy-driven active learning combined with pseudo-labeling using Python and PyTorch.

Game Developer

Unipoly Games

Sep 2021 to Jan 2022

Istanbul, Turkey

- Developed mobile games for iOS and Android using C# and Unity in an Agile environment.

Game Developer

Moraba Games

Feb 2021 to Aug 2021

Tehran, Iran

- Developed hyper-casual mobile games with a focus on rapid prototyping and iteration.
- Optimized ad placement and user experience, increasing ad revenue by ~40%.

PROJECTS

Legal Document Extraction Platform

Technologies: Python, AWS Textract, AWS Bedrock, LangChain, Gradio, PostgreSQL

- Built a full-stack legal document extraction platform to process court awards and populate structured legal templates automatically.
- Integrated OCR and LLM pipelines using AWS Textract and Bedrock to extract structured legal information from unstructured documents.
- Implemented secure user authentication and database-backed storage for accounts, uploaded files, and generated legal records.

Personal Portfolio Website

Technologies: React, Vite, JavaScript, GSAP, GitHub Pages

- Built a modern, responsive personal portfolio website using React and Vite with smooth animations powered by GSAP.
- Implemented dark and light theme toggle with smooth transitions and customizable accent colors.
- Deployed on GitHub Pages with smooth scrolling, project spotlight drawer, and a project spotlight drawer with experience timeline.

Stock Market Prediction

Technologies: Python, Time-Series Analysis, Selenium, BeautifulSoup

- Developed a model to predict stock market trends using historical price data, technical indicators, and time-series analysis.
- Scraped financial news from Investing.com using Selenium and BeautifulSoup, performed sentiment analysis, and correlated sentiment with price trends.

End-to-End ML Pipeline: Churn Prediction

Technologies: Python, Scikit-learn, FastAPI, Docker, MLflow, GitHub Actions

- Built a complete machine learning pipeline including preprocessing, training, and evaluation using Scikit-learn.
- Deployed the system with FastAPI and Docker, tracking experiments with MLflow.
- Integrated CI/CD pipelines using GitHub Actions for automated testing and deployment.

Humming Bird & Flappy Bird RL Agents

Technologies: Unity ML-Agents, Reinforcement Learning, C#

- Developed reinforcement learning agents using imitation learning in Unity ML-Agents.

Face Segmentation Mobile App

Technologies: Unity, TensorFlow, Mobile Inference

- Built a real-time face segmentation mobile application using Unity and TensorFlow.

EDUCATION

University of Calgary

M.Sc. in Computer Science

Calgary, Canada

Sep 2023 to March 2026

Master's Thesis: Segmentation-Based Repair of Unstable Procedurally Generated Game Levels in Angry Birds

University of Tabriz

B.Sc. in Computer Engineering

Tabriz, Iran

Sep 2018 to May 2022

Bachelor's Thesis: Policy Driven Active Learning Combined with Pseudo-labeling and Re-labeling

HONORS & AWARDS

- Faculty of Graduate Studies International Master's Scholarship
- International Graduate Tuition Award
- Mitacs Business Strategy Internship

LEADERSHIP AND SERVICE

Workshop Organizer, Intelligent Narrative Technologies (INT), AIIDE Conference (2024 and 2025)

VP Internal, Computer Science Graduate Society (CSGS), University of Calgary (Jun 2023 to Sep 2025)

Member, Robotics Association of Tabriz University (Sep 2022 to Jan 2023)